CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $(34)_b = 19$ in decimal (a two-digit number in base b), then b equals: 1

(a) 4 **(b)** 5 **(c)** 6 **(d)** 7

2. Weights of 1, 2, 4, 8, ... kg (powers of 2) are available. The least number of weights needed to weigh any whole number of kg up to 31 kg is: 1

(a) 3 **(b)** 5 **(c)** 4 **(d)** 6

3. Which of the following numbers is the greatest? 1

(a) 32_5 **(b)** 1101_2 **(c)** 14_8 **(d)** 16_{10}

4. The number of 1s in the binary form of 77 is: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

5. The Roman numeral for 1994 is: 1

(a) MMIV **(b)** MCMXCV **(c)** MCMXCIV **(d)** MCMXCIII

6. The number that comes immediately after 10111_2 in the binary system is: 1

(a) 11000_2 **(b)** 101110_2 **(c)** 10110_2 **(d)** 11001_2

7. A number written in base 3 ending in the digit 0 is always divisible by: 1

(a) 4 **(b)** 3 **(c)** 10 **(d)** 2

8. If $(25)_b = 17$ in decimal (a two-digit number in base b), then b equals: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 5

9. **Assertion (A):** The binary number 101_2 equals 5 in decimal. 1

Reason (R): In base 2, the value of a number is the sum of each digit times a power of 2, with the rightmost place being 2^0.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The Roman numeral XL represents 40. 1
Reason (R): The Roman symbols V, L and D are never repeated.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the base b if $(23)_b = 13$ in decimal. 2

12. Write 90 in binary and count the number of 1s in it. 2

13. Convert the Roman numeral CXXVII to binary. 2

14. Find the base b if $(52)_b = 37$ in decimal. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 45 g? Explain using binary numbers. 3

16. Add 343_5 and 212_5 (numbers in base 5) and give the answer in base 5. 3

17. How many different numbers can be written using at most 8 digits in base 2? What is the largest of them in decimal? 3

OR

A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 77 g? Explain using binary numbers. 3

18. Add 315_8 and 206_8 (numbers in base 8) and give the answer in base 8. 3

OR

How many different numbers can be written using at most 5 digits in base 3? What is the largest of them in decimal? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** The foundation stone of a school in Nagpur reads MCMLXXXV, and the plaque on the new building reads XXIV. 5
(i) Convert MCMLXXXV to a Hindu-Arabic number.
(ii) Convert XXIV to a Hindu-Arabic number.
(iii) Write 2024 in Roman numerals.
(iv) The school year is 2024; find the age of the school and write it in Roman numerals.

20. In some base b the number $(145)_b$ equals 65 in decimal. Find b and verify your answer by expanding. 5

OR

Subtract $454_8 - 225_8$ in base 8. Verify your answer in decimal. 5

END OF QUESTION PAPER — Total: 40 Marks